MSc Data Analytics

Car Prices Analysis and Prediction Using Machine Learning

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Abstract

Machine learning has evolved very much during the last years, becoming powerful enough to insert in almost every technological sector. Regression models are a small part of machine learning, which is also used in many ways. One of them is price prediction in the market.

There are many different products whose prices can be predicted using regression models. The used cars market is a great example of that, which could help a lot of people take decisions when selling or buying.

We are going to create different regression models to predict the prices of used cars on the market. To do that, we are going to use Python programming language.

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Contents

1. Introduction 1
1.1 Motivation 1
1.2 Project Aims and Objectives 1
1.3 Project Cost, Limitations, Benefits, and Ethical Issues 1
1.4 Methodology 2
2. Background and Literature Review 3
2.1 Introduction 3
2.2 Data Analytics 3
2.3 Machine Learning 3
2.4 Regression Models 4
2.5 Model Evaluation 4
2.6 Related Works 6
2.7 Conclusion 7
3. Requirements and Design 8
3.1 Introduction 8
3.2 Project Requirements 8
3.3 Design 8
3.4 Conclusion 9
4. Data Analysis 10
4.1 Introduction 10
4.2 Chosen Programming Language and Tools 10
4.3 Chosen Dataset 10
4.4 Exploratory Data Analysis 11
4.4.1 Preparation of Data 12
4.4.2 Categorical Variables 13
4.4.3 Numerical Variables 19
4.5 Data Preprocessing and Cleaning 21
4.5.1 Outliers 21
4.5.2 Correlation and Multicollinearity 22

4.5.3 Categorical Variables Encoding 23
4.6 Conclusion 24
5. Model Implementation and Evaluation 25
5.1 Introduction 25
5.2 Chosen Programming Language and Tools 25
5.3 Chosen Machine Learning Algorithms and Evaluation Measures.....25
5.4 Linear Regression 26
5.4.1 Choosing Variables 26
5.4.2 Train 27
5.4.3 Test 27
5.4.4 Experiments 27
5.4.5 Conclusion 28
5.5 Random Forrest 28
5.5.1 Choosing Variables 28
5.5.2 Train 28
5.5.3 Test 29
5.5.4 Experiments 30
5.5.5 Conclusion 30
5.6 Decision Trees 30
5.6.1 Choosing Variables 30
5.6.2 Train 30
5.6.3 Test 30
5.6.4 Experiments 30
5.6.5 Conclusion 31
5.7 Light Gradient-Boosting Machine (LightGBM) 31
5.7.1 Choosing Variables 31
5.7.2 Train 31
5.7.3 Test 32
5.7.4 Experiments 32
5.7.5 Conclusion 33

5.8 Comparison of Results 33
5.9 Conclusion 35
6. Conclusion 36
6.1 Summary 36
6.2 Evaluation of Project Requirements 36
6.3 Project Management 36
6.4 Future Recommendation 36
References 38
Appendix 1: Project Plan 41
Appendix 2: Project Specification 42
Appendix 3: Ethics Review of Project 46

Table of Figures

Figure 2.1: R Squared Mathematical Formulation 5
Figure 2.2: Mean Absolute Error Mathematical Formulation 5
Figure 2.3: Root Mean Squared Error Mathematical Formulation 5
Figure 2.4: Comparison of various Regressions (Madhuri et al., 2019) 6
Figure 3.1: CRISP-DM (Hotz, 2023) 8
Figure 4.1: A sample from the dataset after dropping NA values 12
Figure 4.2: The ten most common regions 13
Figure 4.3: The fifteen most common car manufacturers 13
Figure 4.4: Car condition distribution 14
Figure 4.5: Number of cylinders distribution 15
Figure 4.6: Fuel types distribution 15
Figure 4.7: Transmission types distribution 16
Figure 4.8: Drive types distribution 16
Figure 4.9: Car sizes distribution 17
Figure 4.10: Car types distribution 17
Figure 4.11: Paint colors distribution 18
Figure 4.12: The twelve most common states 18
Figure 4.13: Number of car postings per day 19
Figure 4.14: Price distribution 19
Figure 4.15: Year distribution 20
Figure 4.16: Odometer distribution 20
Figure 4.17: Numerical Variables Statistical Measures 21
Figure 4.18: Correlation Between Numerical Variables 23
Figure 4.19: Multicollinearity 23
Figure 4.20: Dataset After Preprocessing 24
Figure 5.1: Results of Linear Regression Models 28
Figure 5.2: Results of Random Forest Regression Models 29

Figure 5.3: Results of Decision Trees Regression Models 31
Figure 5.4: Results of LightGBM Regression Models..... 32
Figure 5.5: Values of R Squared for all model types 33
Figure 5.6: Values of Mean Absolute Error for all model types 34
Figure 5.7: Values of Root Mean Squared Error for all model types 34
Figure 5.8: Results of All Regression Models 35

Introduction

1.1 Motivation

The use of machine learning has evolved too much during the last few years. More and more scientists are getting interested in working in this field. Professionals and companies tend to use it to make more accurate business decisions and improve their profit. There is no doubt that it is one of the fastest-growing areas of technology. That's because there are many different sectors that machine learning is applied, and a lot of different techniques to use it.

The most common way of using machine learning is by creating models that can later be used to make predictions. A very important type of model that can be created is a regression model (Colin & Michael Lewis-Beck, 2016). The reason regression models are important is that, given some input, they can predict continuous values and return an exact result. They could predict the earnings of a company, the height of a child, or the selling price of a car.

The last example is what this project is about. Creating a regression model that can predict the price of used cars as accurately as possible. Different regression model types will be used, to find out which is the best to use for this purpose.

The reason this project idea was chosen is that it's very interesting to use data to predict the price of an asset on the market and I would like to see how accurate the models can be. The used cars market is a very special type of market because except from professionals there are also many individuals selling cars, and this results in many fluctuations in the price of a car. Both people selling a car and people who want to buy a car could use the results of this research to help them get a better idea of what the price of a car should be.

My work can be helpful for future researchers who try to perform similar research. It can be compared and combined with other similar projects to improve the results in the future.

1.2 Project Aims and Objectives

This project aims to create the best regression model possible, given the dataset that is going to be used. To achieve that, a complete data analysis will first have to take place. Then, different regression model types will be tested. Several versions of each type are going to be created, using different attributes, and the best of each will be used. Finally, the different regression types model's results will be compared, to identify which one is the best.

1.3 Cost, Limitations, Benefits, and Ethical Issues

Regarding the costs of the project, all the necessary research and work was performed remotely from my home, using my pre-existing computer, so there were not any research-related costs. The tools-programs used for the coursework were only Python and Microsoft Word. Python is a free, open-source programming language, while Microsoft Word has been provided by the university for free. So, there was totally zero cost in performing the coursework.

There were two basic limitations related to the project. The first was the difficulty of finding the dataset. While there were several similar datasets regarding car sales data available on the internet,

most of them were in very bad condition or did not contain the information required for this project. So, a lot of searching was required to find the final dataset. The second limitation was the limited time I had while doing the coursework. Due to working full-time during the period of the project, my available time to study was very limited compared to when I was unemployed. So, very good time management was needed.

Regarding the benefits of this project, other researchers with an interest in similar projects can use it as a starting point and try to improve its results, or they could use it to compare their results. It could be useful for other projects involving not only car price prediction, but any type of machine learning project including regression models.

There were no ethical issues with performing the project. That's because the dataset that was used is a publicly available dataset found on Kaggle.com.

1.4 Methodology

The methodology used to perform the tasks required to achieve the project's requirements is similar to the Cross Industry Standard Process for Data Mining (*CRISP-DM*) method (Schröer et al., 2021). This is a standard process that is used for data mining/analytics. Based on this method, there are six phases for such projects:

1. Business understanding: Understanding the business needs.
2. Data understanding: Understanding what data we have and what they contain.
3. Data preparation: Organizing data for modelling.
4. Modelling: Choosing the modelling techniques.
5. Evaluation: Choosing the best model based on results.
6. Deployment: Accessing the results.

Specifically for this project, the first step involves understanding how the car market works and all the car-related information that exists in the dataset. The second step is about fully understanding the variables in the data and performing the necessary changes to the dataset. The third step involves choosing the models that will be used and then training the models using the data. The final step is evaluating the results, comparing the different models, and choosing the best model created that can be used for car price predictions.

2. Background and Literature Review

2.1 Introduction

This chapter is separated into two different parts. In the first part, background, all the areas of computer and data science that are needed to perform this project are going to be discussed. The terminology of each of them will be explained, to understand why it is useful to create car price prediction models. In the second part, the literature review, some similar projects will be mentioned to see what results they had.

2.2 Data Analysis

Data analysis is described a little differently based on the experts/organizations that are using it. A general definition would be the combination of understanding, processing, and cleaning data, to extract important information that may be hidden inside them (Lewis-Beck, 1995). This kind of information can be used by experts, businesses, and scientists to increase their profits, make better business decisions, avoid high business risks, etc. We can easily understand the advantage of the use of data analysis related to the past when this kind of information was not available.

To understand its value, as an example, imagine if we could make a decision in our everyday life, knowing exactly what the results of similar decisions were for other people in the past. This would give us the power to know what the most possible outcome is by making that decision and help us make a better final choice. This is similar to the way that businesses and data experts use data analysis.

There are numerous areas where it can be applied, and this is another reason it is so important. Some of these areas are healthcare (Raghupathi & Raghupathi, 2014), finance (Ravi & Kamaruddin, 2017), banking (Srivastava & Gopalkrishnan, 2015), transportation (Chowdhury et al., 2017), logistics (Ben Ayed et al., 2015) manufacturing (Sahoo, 2021), insurance (Boobier, 2016), marketing (Ducange et al., 2018), education (Nguyen et al., 2020), security (Cárdenas et al., 2021), and fraud detection (Bănărescu, 2014). It is one of the newest technologies and is expected to be used in almost any sector and by even more businesses in the future.

Specifically for this project, data analysis will be the first task done before creating car prediction models. It is necessary because it will help us understand the data and do any important preprocessing needed. This way, data will be transformed into the desired form, making it able to be used for modelling.

2.3 Machine Learning

Machine learning is one of the fastest-evolving areas of data science. It is a mix of artificial intelligence and computer science that uses algorithms on data to create models that can do special things like making predictions (Zhou, 2016). What's special about it is that these algorithms try to mimic the way in which humans acquire knowledge, steadily enhancing the precision of the prediction models, without a lot of human input.

Using statistical methods, algorithms undergo training to perform predictions and reveal crucial information in data mining projects. These insights are used by businesses, experts, and scientists to improve decision-making and key growth metrics for companies.

Different types of machine learning are used based on the user's purpose. The machine learning category that is going to be used for this project is called supervised learning (Cunningham et al., 2008). This type of machine learning is used when the dataset is labeled, to train models that can make predictions or classify data.

This is exactly what we also want to do in this project. We will use supervised machine learning, after the data analysis is complete, to create machine learning models. The type of models that will be created are called regression models.

2.4 Regression Models

Regression models are used to create a function that describes how the independent variables affect the dependent variable (Colin & Michael Lewis-Beck, 2015). This function can then be used to make predictions for the target variable. To achieve that, the regression algorithm must be trained on a set of test data. This is exactly what we want to do in this project and that's why regression models were used.

Many different types of regression algorithms can be used based on the occasion. In this project, the regression algorithm types that will be used are linear regression, random forest regression, decision trees regression, and light gradient-boosting machines.

Linear regression is used to describe the relationship between the dependent and independent variables in a straight line (Xin, 2009). It is probably the most common regression type used because it is the easiest to understand and observe in real life.

Random forest uses ensemble methods to perform regression (Pavlov, 2000). These are techniques that use multiple algorithms combined to create a model. This enables them to make more accurate predictions than using only a certain model.

Decision trees create regression models in the form of tree structures. They split the dataset into smaller and smaller parts until a tree is created, with decision and leaf nodes (Kingsford & Salzberg, 2008). The decision nodes have more than one branch which represents the values of the tested attribute. The leaf nodes represent the value of the target numerical variable.

Light gradient-boosting machines is a gradient-boosting framework that is based on decision trees (Aziz et al., 2022). This framework can achieve high accuracy in predictions and reduced memory usage because they have a type of automatic feature selection and focus on larger gradients.

2.5 Model Evaluation

The last and very important part of the project is model evaluation. It is the process of applying different statistical metrics for the evaluation of the models. It helps data scientists understand how well a model performs and identify any strengths or weaknesses. Without evaluation, we would not be able to understand the quality and accuracy of the models or compare them and understand which performs the best.

In this project the model evaluation will take place after the machine learning section, to evaluate the models' results, compare them and choose the best model. The metrics that will be used are the r-squared, the mean absolute error, and the root mean squared error (Armin et al., 2013).

R-squared is a statistical metric that represents how much of the variance of the target variable is explained by the variance in the independent variable(s) (Miles, 2005). So, a high r-squared value means that the model is performing well, as a lot of the variance of the dependent variable is explained by the variance in the independent variable(s). The value of r-squared is between 0 and 1, and a value above 0.50 is considered acceptable.

Figure 2.1: R Squared Mathematical Formulation.

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

Mean absolute error is the average level of difference between the predicted and actual values of an observation. Absolute errors are the differences between the actual and predicted values, and their mean gives a complete idea of the entire group of observations. Its value can range from 0 to infinity. Smaller differences between predicted and actual amounts indicate better model performance, so the value of mean squared error is better as it gets closer to 0.

Figure 2.2: Mean Absolute Error Mathematical Formulation.

$$MAE = \frac{1}{N} \sum_{i=1}^{N} |\, y_i - \hat{y}_i|$$

Root mean squared error is the root of mean squared error. It measures how near a set of data points is to a regression line. In other words, it calculates the mean of the squares of the errors (residuals), which is the average of the squared difference between the estimated and the actual values. The reason that the root of the mean squared error is used is that it provides a greater interpretation of the results, making it easier to understand how the model performs. The value of root mean squared error can reach from 0 to infinity. As the number of errors decreases, the value of root mean squared error gets closer to 0, so the closer it is to zero, the better performance the model has (Chai & Draxler, 2014).

Figure 2.3: Root Mean Squared Error Mathematical Formulation.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2}$$

2.6 Related Works

There are two works that will now be mentioned because they are similar to this project. The first one is from Venkatasubbu and Ganesh (2019), who published a paper in the International Journal of Engineering and Advanced Technology (IJEAT) named Used Cars Price Prediction using Supervised Learning Techniques. The main objective of this paper was to predict the retail price of a used car using different supervised model techniques.

To achieve that, three different prediction models were used and needed to be compared regarding their level of accuracy, lasso regression, multiple regression, and regression tree. The data was collected from the 2005 Central Edition of the Kelly Blue Book which has 804 records of 2005 GM cars, whose retail prices had been calculated.

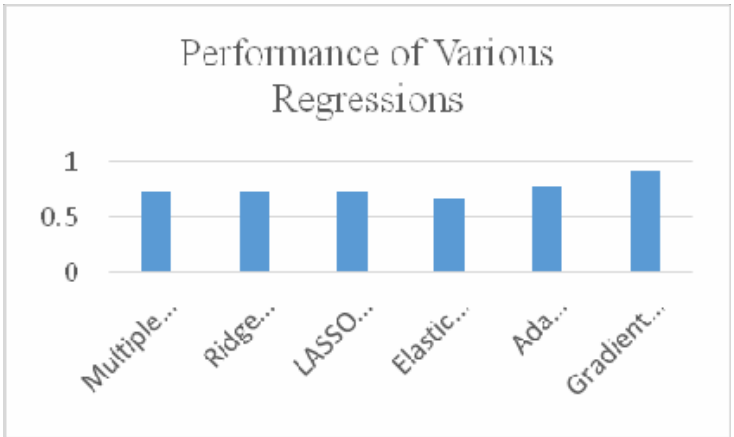
The result of their research was that the prediction error rate of all the models was well under 5% of error, but on further analysis, the mean error of the regression tree model was found to be more than the mean error rate of the multiple regression and lasso regression models. Even though for some seeds the regression tree had better accuracy, its error rates were higher for the rest. Also, the error rates in multiple regression models and lasso regression models were not significantly different from each other.

The second work from Madhuri et al. (2019), was a report published for the IEEE 6th International Conference on smart structures and systems regarding house price prediction using regression techniques. Their objective was to forecast the coherent house prices for non-householders based on their financial provisions and their aspirations.

To do that, they analyzed the foregoing merchandise, and fare ranges and forewarns developments. Speculated prices were also estimated. The paper involves predictions using different regression techniques like Multiple linear, Ridge, LASSO, Elastic Net, Gradient boosting, and Ada Boost Regression. House price prediction on a dataset was done by using all the mentioned techniques to find out the best among them.

The result of their research was that out of all the regression techniques, the gradient boosting algorithm had the highest accuracy (score), regarding house price predictions.

Figure 2.4: Comparison of various Regressions (Madhuri et al., 2019).



2.7 Conclusion

In this chapter, data analysis, the terms machine learning, regression models, and model evaluation were explained to provide background information for this project. A related work was also explained, which is very close to this project.

3. Requirements and Design

3.1 Introduction

In this section, the requirements and the design of the project will be discussed. This will help us understand what exactly is required in the project and how we will achieve this result.

3.2 Project Requirements

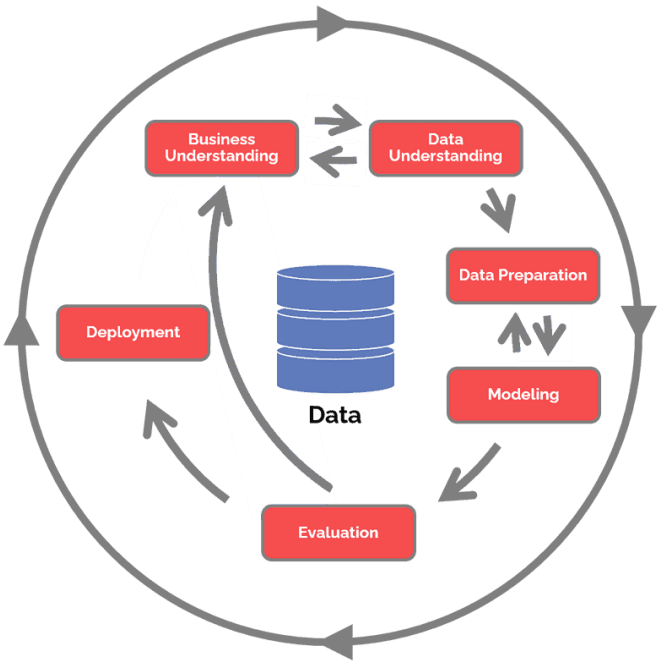
The main requirement of this project is to create a regression model that can predict the price of used cars. Before reaching the final model, multiple regression models must be created, using different regression algorithm types. The performance of these models has to be calculated and their results must be compared, to finally keep only the best-performing model.

The secondary requirement of the project comes before the creation of the regression models. It is required to perform extended data analysis, including exploratory analysis and preprocessing of the dataset. This requirement is mandatory because it is the best way to understand the data and prepare for the modelling part.

3.3 Design

As mentioned in the '1.4 Methodology' section, the Cross Industry Standard Process for Data Mining (*CRISP-DM*) is the design method for the project, since it is very close to how the project was approached. Business understanding is the first step because we first need to understand how the used cars market works. Next, data analysis is needed to perform the data understanding and data preparation for the project. Modelling and evaluation are the most important parts of the project where the models will be created and the best one will be identified. The deployment stage will not take place in the project.

Figure 3.1: CRISP-DM (Hotz, 2023).



Another part of the design section of the project is the tools and technologies used. To perform all the analysis and modelling tasks of the project, Python programming language was used. To write Python code, a platform or program is needed, and that is Jupyter Notebook.

3.4 Conclusion

Now that the requirements and design of the project have been discussed, we are going to move to the technical part of the project. Starting with the data analysis.

4. Data Analysis

4.1 Introduction

Data Analysis is called the process of analyzing, inspecting, and making transformations on the data, to drain useful information from it. Numerous different techniques and procedures can be performed for data analysis, which has to do with data exploration and cleaning.

Before moving to any machine learning technique, it is crucial to apply data analytics methods to understand the data and prepare it for modelling. Understanding the data means getting to know exactly what features the dataset contains and what values these features can have. This is helpful when deciding how to use the data and what to do with it. The next step is preparing the data for modelling, which means making any useful transformations to the features or dropping the ones that are not useful. Without performing such actions, it is not possible to move on and use machine learning algorithms.

For the purposes of this project, in this section, we are going to discuss which tools will be used to perform the analysis, why this dataset was chosen, what it contains, and all the data analysis-related tasks that will be performed step by step.

4.2 Chosen Programming Language and Tools

The programming language that will be used for the data analysis part is Python. It is one of the most widely used tools for data analysis, and that's because it offers many different functionalities related to other tools. This is achieved by using different libraries, which are like more specific tools used in Python, each one for a more specific cause.

The libraries that will be used for the Data Analysis part are Numpy, Pandas, Matplotlib, and Seaborn.

- Numpy is one of the core libraries when using arrays and statistics/mathematics.
- Pandas is the most important library when working with datasets because it is used to perform most of the analysis and transformation tasks.
- Matplotlib and Seaborn are used to create visualizations and graphs to get a better understanding of the data.

To run the Python code, the Jupyter Notebook application will be needed. Jupyter Notebook is one of the most famous applications to write and write code, especially for Python, which was ideal for our project.

So, before starting the data analysis and machine learning tasks, all the necessary tools needed to be installed. For that reason, Python 3.10.7 was the first one. Then, it was the turn for all the necessary libraries, using the 'pip install *library_name*' command on the command line. Finally, Jupyter Notebook was also installed and ready to use.

4.3 Chosen Dataset

The dataset that is used is called the 'Used Cars Dataset'. It is a publicly available dataset found on Kaggle.com, a well-known website for datasets, which contains data about car sale listings in the

United States of America. It has a lot of information about used cars, some of which can be very useful for creating price-predicting models. More specifically, the dataset has 26 attributes and 426.880 records of data. It is a CSV file that is easy to handle. Let's explore what each attribute is about.

The dataset's attributes are:

- id: the unique id of the car that is posted for sale
- url: the url of the car sale listing
- region: the region in the United States of America where the car belongs to
- region_url: the url of each region
- price: the selling price of the car (the attribute we are going to predict)
- year: the year that the car was manufactured
- manufacturer: the brand that manufactured the car
- model: the manufacturer's model of the car
- condition: the current condition of the car
- cylinders: the number of cylinders the car has
- fuel: the type of fuel of the car
- odometer: the number of miles the car has covered
- title_status: the title status of the car
- transmission: the transmission type of the car
- VIN: the identification number of the car in the United States
- drive: the type of drive of the car
- size: the size type of the car
- type: the generic type of the car
- paint_color: the color of the car
- image_url: an image of the car
- description: the description included in the car sale listing
- county: the county where the car belongs to
- state: the state in the United States of America where the car belongs to
- lat: the latitude of the car listing
- long: the longitude of the car listing
- posting_date: the date that the car listing was posted

The main reason that this dataset was chosen is that it contains so many attributes, which cover the most important parameters when deciding the price of a used car. It also has data on 426.880 different car sale listings. The high number of attributes and records gives us the advantage of being able to make a lot of modifications and tests, to finally get the best results. Of course, not all attributes and records will finally be used, but this will be discussed later.

4.4 Exploratory Data Analysis

The most important thing before moving to machine learning is understanding the data we are going to use. To achieve this, some exploratory data analysis (EDA) was done. This kind of data analysis focuses on analyzing data to determine their main characteristics, using statistical methods and data visualizations.

4.4.1 Preparation of Data

Before starting the analysis, all necessary libraries need to be imported (numpy, pandas, seaborn, and matplotlib). The dataset is loaded and named 'df'. To get a first idea of what the dataset contains, how many records, and what kind of columns there are, the commands df.head(), df.info(), and df.shape() are executed.

With just a glance, it is noticed that some variables ('id', 'VIN', 'url', 'region_url', 'title_status', 'image_url', and 'description') are not going to be useful in making predictive models because they do not contain information about the characteristics of the cars. Variables 'lat' and 'long' contain information about the exact position of the car which will not be needed for the purpose of this project. Variable 'county' is indicated by the source of the dataset that contains only null values. So these variables are dropped from the dataset, and only 16 columns are left.

The next step includes a check for duplicate records. These are records that exist more than one time in a dataset and are possibly generated by mistake during the creation of the dataset. It is very important to exclude them because they may cause noise and lead the models to overfit some values. So, any possible records are dropped. After checking the number of rows in the dataset again, no records were dropped. That means there were no duplicates.

Another very important step is checking for NULL (or NaN or Na) values. When the value of an attribute is set as NULL it means that this information is missing, and the row is usually dropped. Rows that have a NULL value in a column may contain other useful information in other columns, so it depends on the person that handles the data and the purpose of the project if these records are going to be dropped. In our case, any rows including NULL values are dropped. That's because we have a very satisfying number of records in the data and even if we remove a big number of records, we will still have all the information new need. Using the 'df.dropna' command, all records including NULL values are removed.

After these changes, the dataset has 79.193 records and 16 columns. There are no duplicated records and no NULL values. 'Figure 4.1' shows how the dataset currently looks.

Figure 4.1: A sample from the dataset after dropping NA values.

region	price	year	manufacturer	model	condition	cylinders	fuel	odometer	transmission	drive	size	type	paint_color	state	posting_date
auburn	15000	2013.0	ford	f-150 xlt	excellent	6 cylinders	gas	128000.0	automatic	rwd	full-size	truck	black	al	2021-05-03T14:02:03-0500
auburn	19900	2004.0	ford	f250 super duty	good	8 cylinders	diesel	88000.0	automatic	4wd	full-size	pickup	blue	al	2021-04-29T17:19:18-0500
auburn	14000	2012.0	honda	odyssey	excellent	6 cylinders	gas	95000.0	automatic	fwd	full-size	mini-van	silver	al	2021-04-27T12:20:01-0500
auburn	22500	2001.0	ford	f450	good	8 cylinders	diesel	144700.0	manual	rwd	full-size	truck	white	al	2021-04-26T11:15:36-0500
auburn	15000	2017.0	dodge	charger rt 4dr sedan	excellent	8 cylinders	gas	90000.0	automatic	rwd	mid-size	sedan	grey	al	2021-04-24T18:39:59-0500

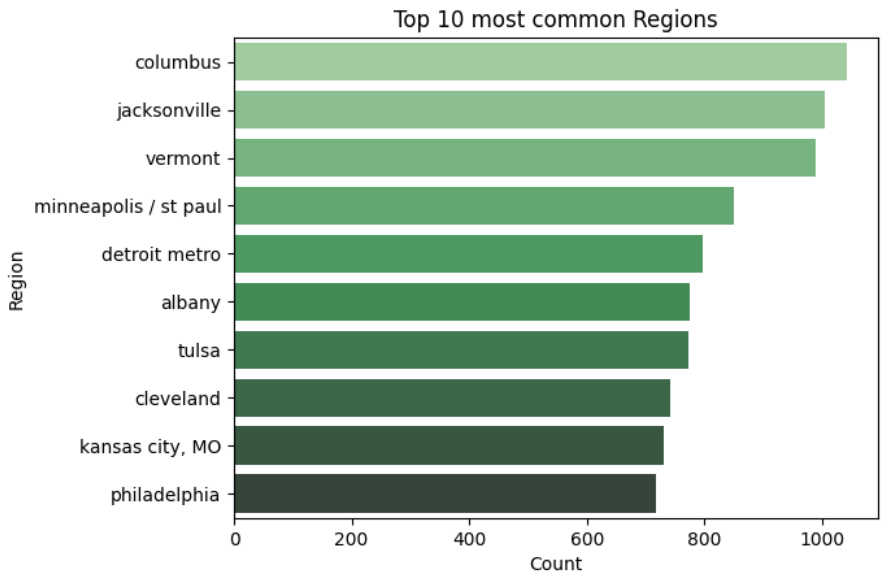
At this point, we are ready to start analyzing each of the 16 columns further. This will help us understand better what information each column has, so we can keep only the rows and columns we consider useful. For that purpose, different graphs will be presented, and the 'df.value_counts' command will be used a lot to check what values each attribute can take.

There are two types of variables in the dataset, categorical and numerical. Categorical variables are the ones whose data are divided into different groups. Numerical variables take values that are described as a number. We are going to start with the categorical ones.

4.4.2 Categorical Variables

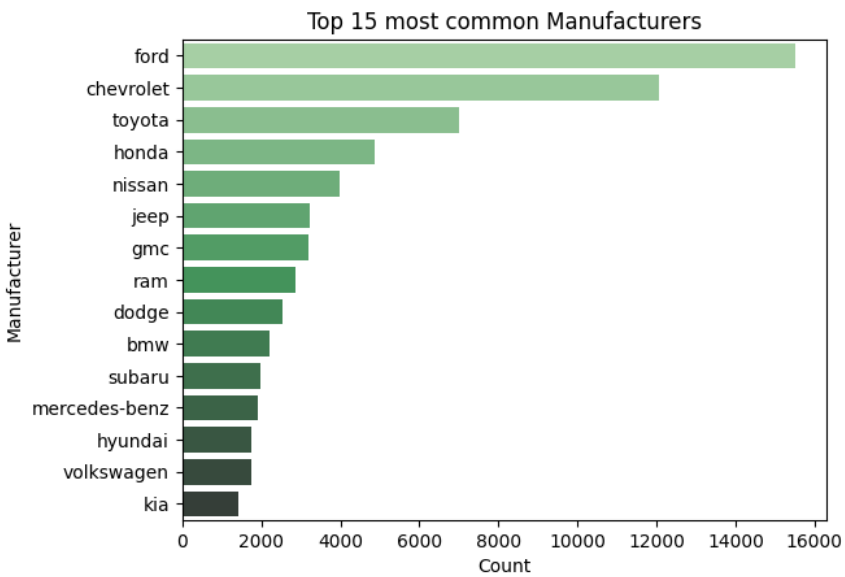
The first categorical attribute 'region' consists of 403 different regions. All of them have valid values, so no changes will be made. The number of observations per region goes from 1 in 'southwest TX' to 1043 in 'columbus'. Not all of them can be represented, so only the 10 most common regions are in 'Figure 4.2'.

Figure 4.2: The ten most common regions.



The second attribute 'manufacturer' contains 41 different car manufacturers. Again, all values are valid, and no manipulations are needed. The number of observations per manufacturer goes from 5 for 'aston-martin' to 15.515 for 'ford'. Only the 15 most common ones are in 'Figure 4.3'.

Figure 4.3: The fifteen most common car manufacturers.

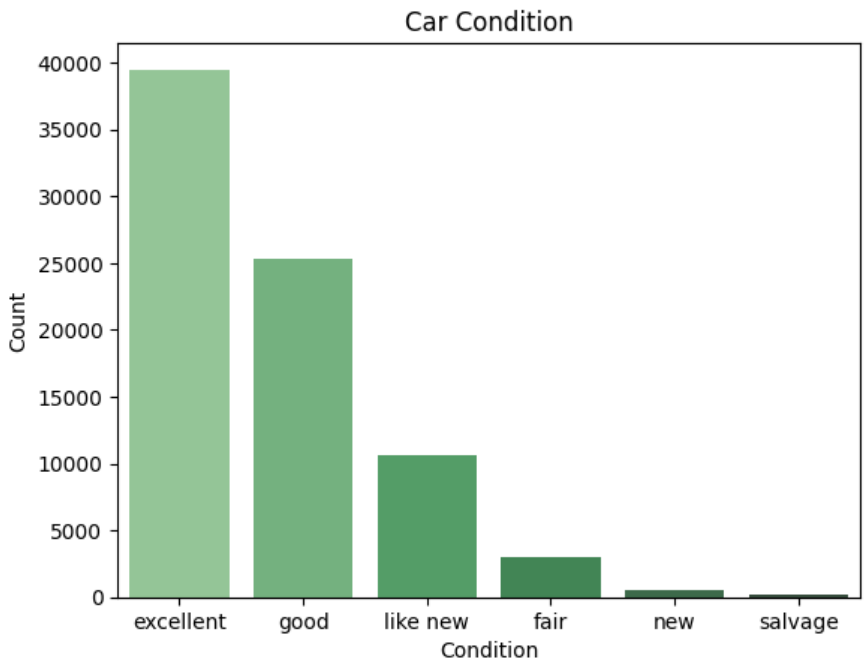


The next attribute in the dataset is 'model' which contains 9.539 different car models. After checking the different values this attribute can take, it was decided not to be used, because:

- There are too many different car models (9.539).
- Some records have very simple kinds of values (i.e. '3', '1500') which can belong to different manufacturers. This can be misleading for the models.
- 'manufacturer' attribute has a similar kind of information, so using 'model' would not add much different information for the models.

Next is the 'condition' attribute. It has 6 different values which we can see in 'Figure 4.4'. They are all in good condition, providing useful information about the current condition of the cars. We notice that most cars appear to be in excellent or good condition.

Figure 4.4: Car condition distribution.

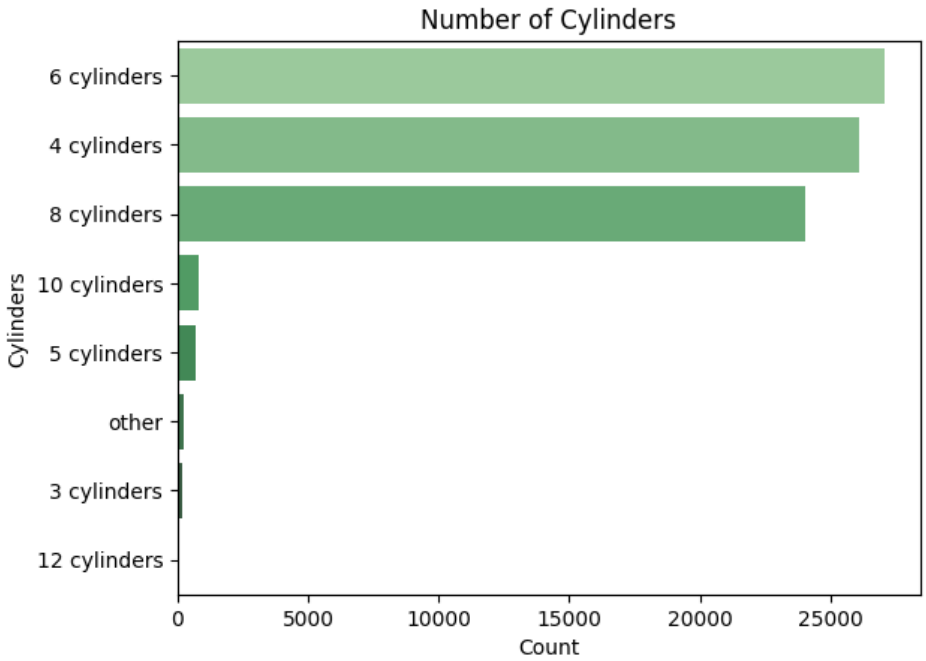


Attribute 'cylinders' has 8 different values as we can see in 'Figure 4.5'. While all of them are in good condition, it was decided not to use the rows which take the value 'other'. That's because:

- This value indicates that we do not know the number of cylinders of the car, so it is like a NULL value.
- There are only 250 such records, so not a lot of information will be lost.

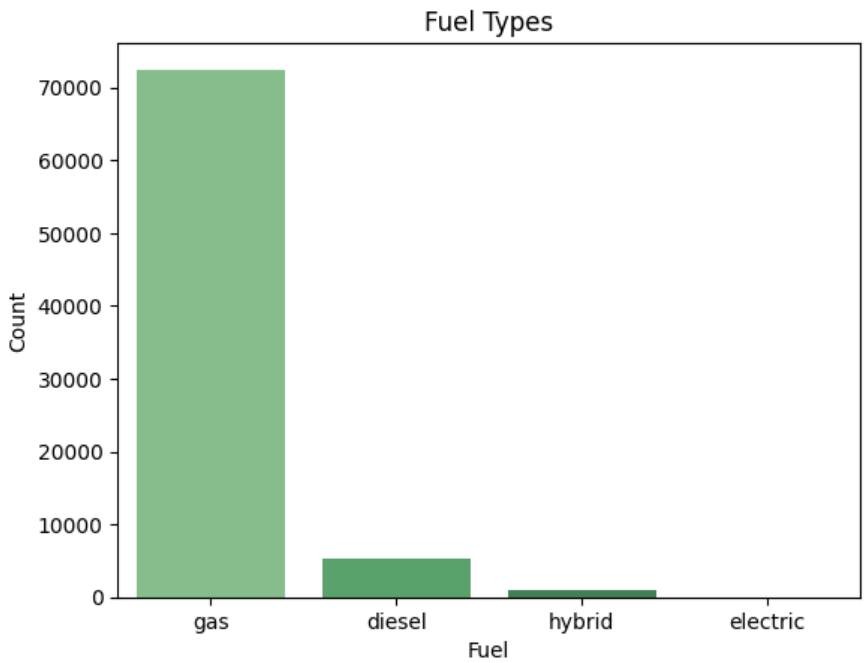
We notice that most cars have 4, 6, or 8 cylinders.

Figure 4.5: Number of cylinders distribution.



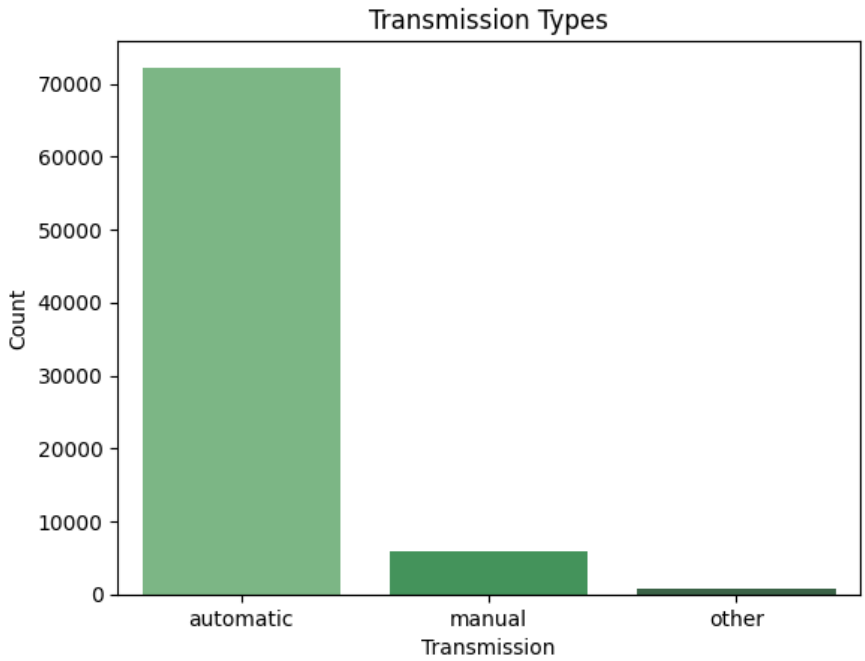
Next is the attribute 'fuel', which takes only 4 different values. Again, they are in good condition, but it was decided not to use the ones that take the value 'other', for the same reason as in the 'condition' variable. So, 95 such records are dropped. We notice in 'Figure 4.6', that most cars have gas as a fuel type.

Figure 4.6: Fuel types distribution.



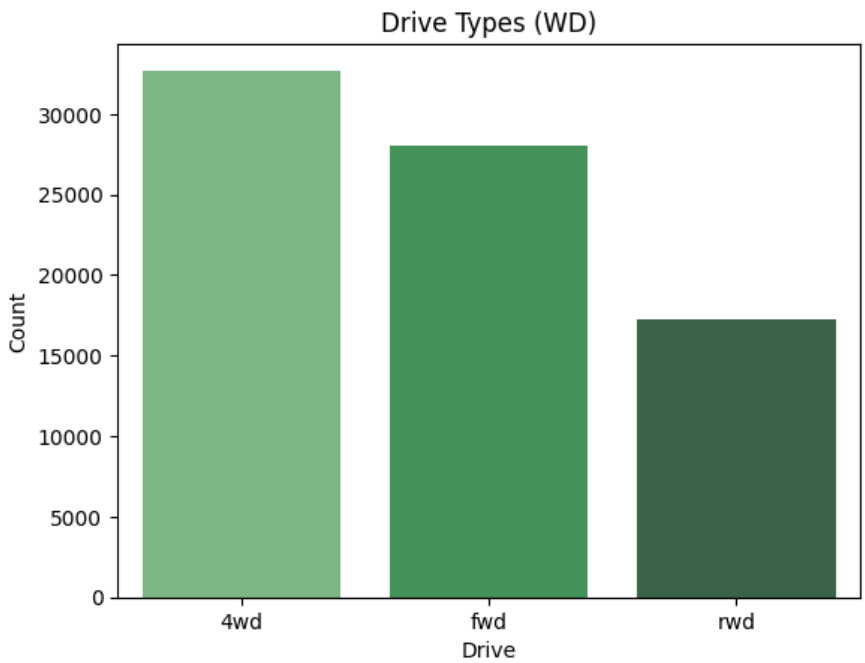
Regarding variable 'transmission', the same process is followed. It takes only 3 different values, one of which is 'other' that will not be used. So, 834 more records for which we do not know the transmission type are dropped. We can see in 'Figure 4.7' that most cars are automatic.

Figure 4.7: Transmission types distribution.



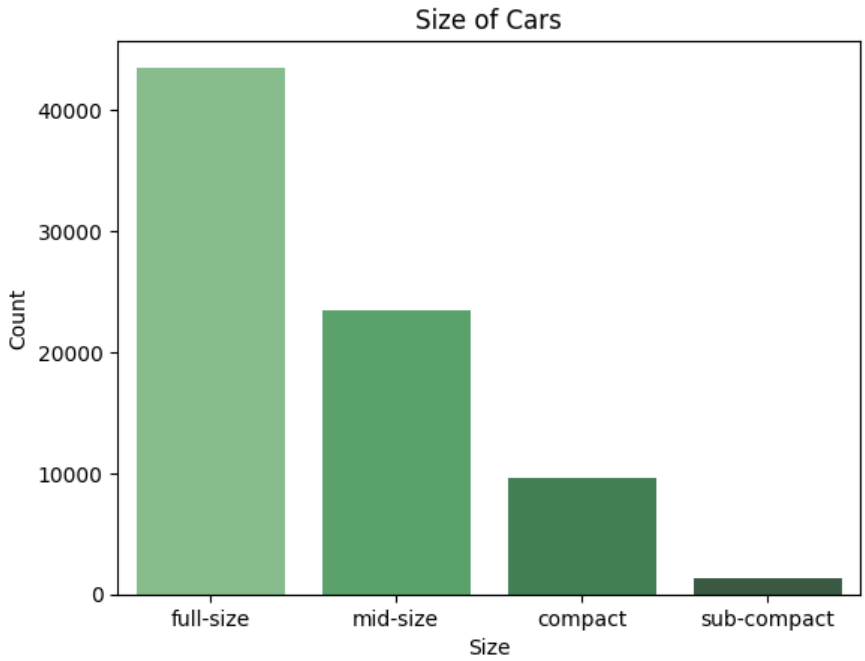
As we can see in 'Figure 4.8', attribute 'drive' takes only 3 different values, which are in perfect condition, so no changes are made. The 3 values are almost evenly distributed.

Figure 4.8: Drive types distribution.



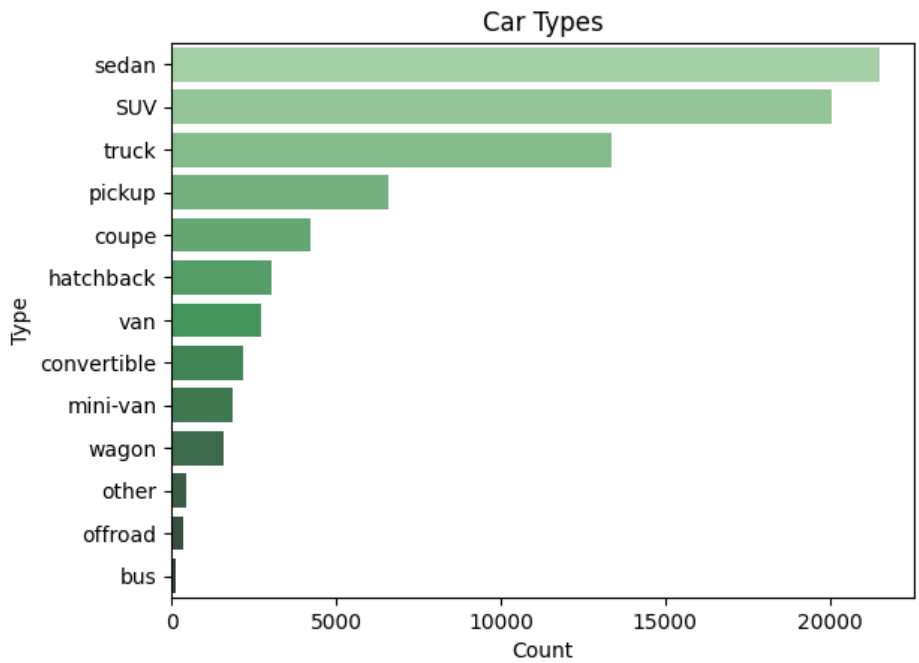
The 'size' attribute is also in very good condition, as it takes only 4 different values which are all in a good, understandable condition. In 'Figure 4.9' we can see that most cars are full-sized.

Figure 4.9: Car sizes distribution.



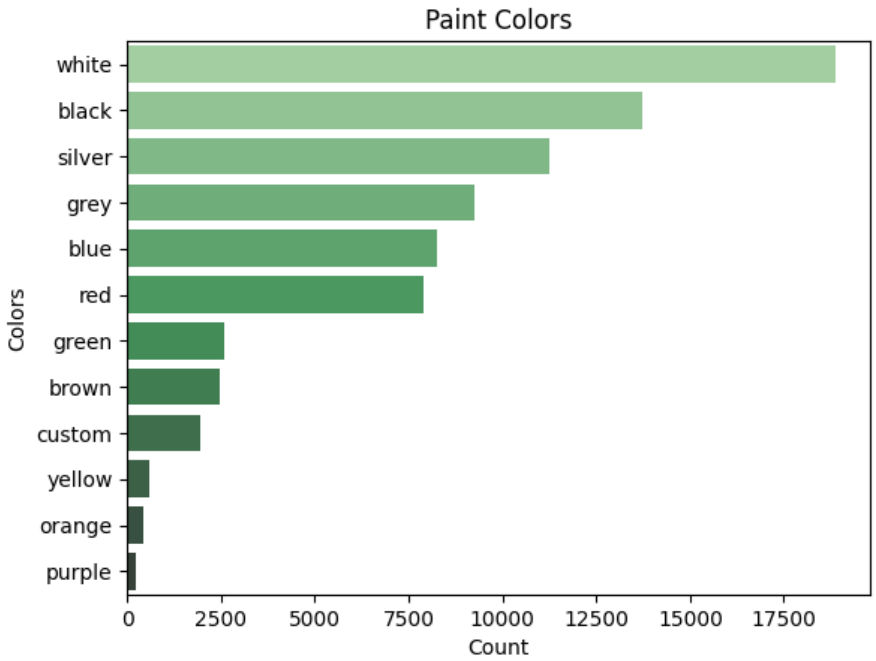
Following up is the 'type' variable, which has 13 different values. As we noticed in some previously mentioned variables, while all values are in good condition, rows taking the value 'other' will be dropped. This means that 464 more rows are gone. In 'Figure 4.10' we notice that most cars are sedans or SUVs.

Figure 4.10: Car types distribution.



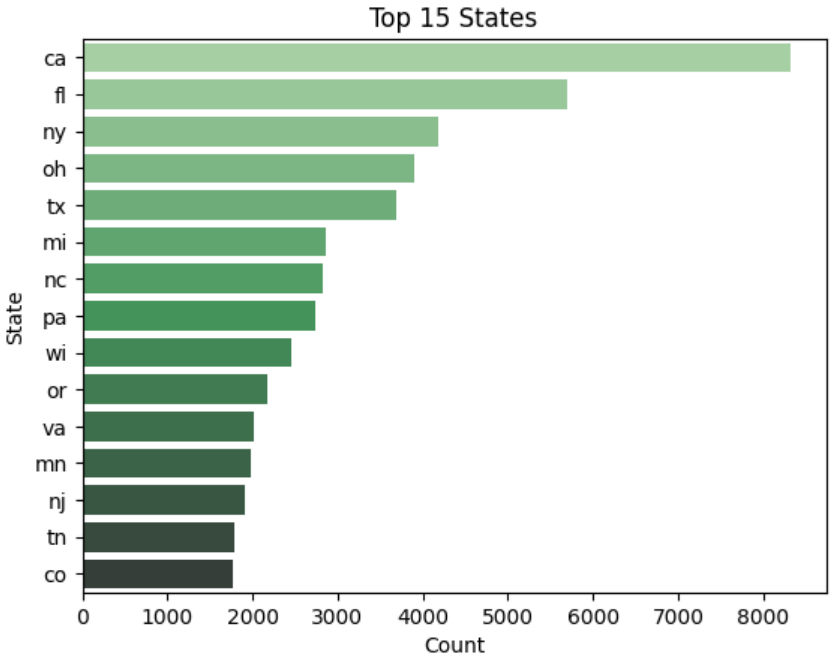
The next attribute is 'paint_color'. As we can see in 'Figure 4.11', it takes 12 different paint values, all of which are in good condition, so no changes are made. We notice that most cars are white or black.

Figure 4.11: Paint colors distribution.



The next variable we have is 'state', which takes 51 different values, all in a readable form. In 'Figure 4.12' we can see that most cars belong to the states of California and Florida.

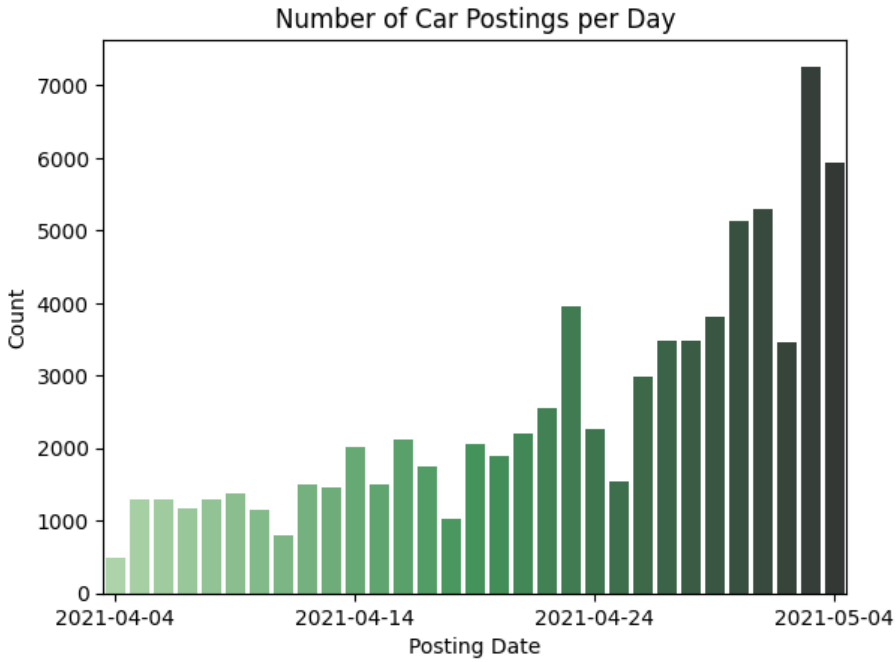
Figure 4.12: The twelve most common states.



The last categorical variable in the dataset is 'posting_date'. First, this variable contains both the date and time of the posting. The time is too specific information that we will not need, so only the date is kept. The posting dates range from 05/05/2021 to 04/05/2021. It is a date-type variable but because we only have a few dates, we are treating it like a categorical one. We can see in 'Figure 4.13' that as time passes until the start of May, more and more cars are being posted for sale. There might be a seasonality reason behind this trend, i.e., as the summer is closer, more people would like to sell their

car and buy a new one. The number of car postings correlates positively with the posting date, and this is something that might be useful for our models. The problem is that we cannot be sure of this theory since the period is very small. No changes are made for now.

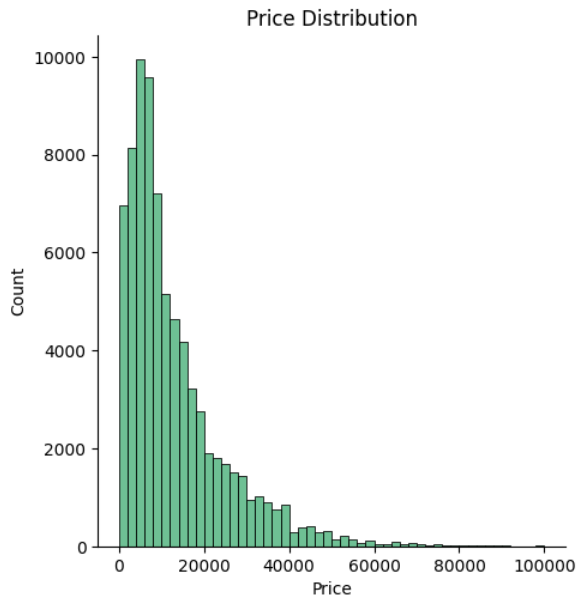
Figure 4.13: Number of car postings per day.



4.4.3 Numerical Variables

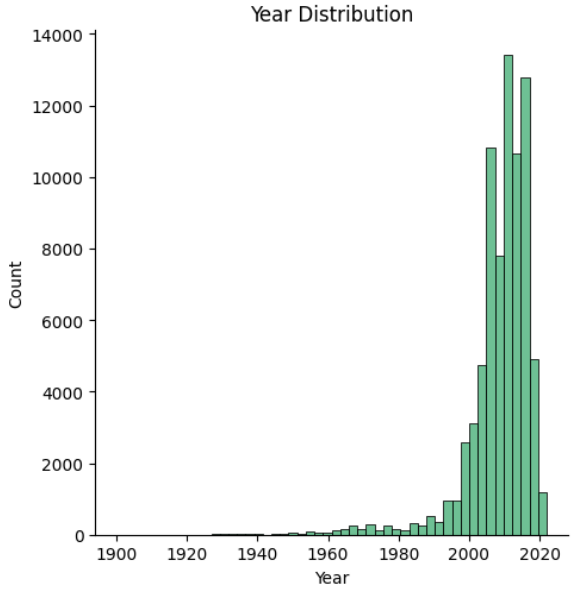
Now it is time to discuss the numerical variables, which are only two, 'price' and 'odometer'. Price is the variable that we want to predict. Using the 'df.describe' command we can see that prices range from 0$ to 3,736,928,711$. The mean price is 80,891$, while the median is only 9,000$. This is because of some extreme values in price. Because of these extreme values, only prices under 100,000$ are represented in 'Figure 4.14'.

Figure 4.14: Price distribution.



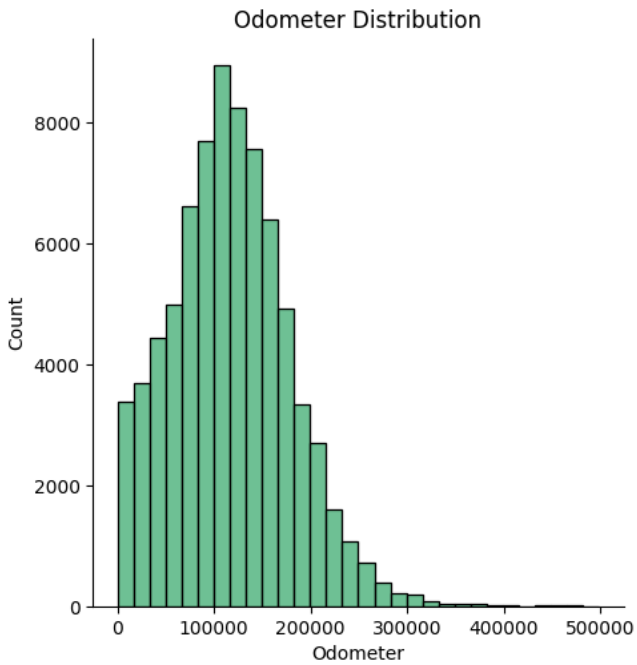
The ‘year’ variable is not completely numerical, since it refers only to a year value, but for the purpose of the project, we are going to treat it as numerical. It takes 102 different values, ranging from a minimum of 1900 to a maximum of 2021. Its values are in good form and no changes are necessary. As most people would expect, we can see in ‘Figure 4.15’ that most cars that are listed for sale were made after 2000.

Figure 4.15: Year distribution.



The last attribute of the dataset is ‘odometer’. Again, using the ‘df.describe’ command we see that odometer values rank from 0 to 10.000.000 miles. The mean is 124.849 miles while the median is 114.692 miles. Because there are some very high values in the ‘odemeter’ variable with just a few occurrences, only cars with less than 500.000 miles of odometer are used in ‘Figure 4.16’.

Figure 4.16: Odometer distribution.



After these small changes were applied while checking the values of the variables, there are now 15 columns and 77.550 records left, without duplicates, NULL, or unexplainable values, which is a very satisfying number. Now that we have a good knowledge of the dataset, we can move to the preprocessing part of the analysis.

4.5 Data Preprocessing and Cleaning

4.5.1 Outliers

Outliers are observations in data whose values differ a lot from the dataset's average. They are excluded to avoid causing noise to the data. The level of outliers that are excluded though, is subject to the person performing the analysis, based on the project's needs. Such values may exist in the numerical and date variables.

To identify outliers, we can plot the data or use statistical measures, like the IQR (Interquartile Range). In our case, after plotting the data (like we did before), we identified there are outliers in some variables. 'Figure 4.17' below shows some statistical measures that also show the same. These outliers will need to be excluded or they will cause problems when creating the models later.

These variables having outliers are the numerical ones ('price', 'odometer', and 'year'). After all the exploratory analysis we performed, with the good knowledge of the variables we have, it was decided the level of outliers that is going to be chosen manually (not using IQR) for each variable.

Figure 4.17: Numerical Variables Statistical Measures.

	price	year	odometer
count	77550.000000	77550.000000	77550.000000
mean	80890.990264	2008.514778	124848.846319
std	14022145.675521	9.859101	241656.153483
min	0.000000	1900.000000	0.000000
25%	4995.000000	2006.000000	74000.000000
50%	9000.000000	2011.000000	114692.000000
75%	17500.000000	2014.000000	156000.000000
max	3736928711.000000	2022.000000	10000000.000000

The 'price' variable has a mean of 80.891$ and a median of just 9000$, while the 3rd quartile is 17.500$, and, as we can see in the 'Price Distribution' graph, most prices are under 20.000$. These numbers indicate the existence of some extreme outliers since they can increase the mean so much and create such a big difference from the median. These extreme values need to be excluded, and as mentioned before, this is going to happen manually.

So, prices that exceed the amount of 50.000$ are dropped, an amount which was chosen after testing many tests. We could say that cars that have a selling price of over 50.000$ are considered luxury cars and the criteria for the calculation of their price might be different. These cars are a minority, and the prediction of their prices can be done separately in a different project. Cars with a price of 0$ are also

dropped because that means their selling price is not available and these records can be completely misleading. The total number of records dropped after this procedure is 1.190.

Next is the 'odometer' attribute. We can see in the 'Odometer Distribution' graph that most records have values under 300.000 miles. From the table above we understand that there is probably a small number of extreme values since the max is 10.000.000 miles while the 3rd quartile is only 156.000 miles, and the mean is 123.849 miles.

So, records with 'odometer' values over 300.000 miles are dropped. Also, some records with 0 miles values are removed, since that means that we are referring to a new car, not a used one. So, a total of 581 records are gone.

The last variable we are checking about outliers is 'year'. It is clear from the 'Year Distribution' graph that most cars were made after 2000. Even though, looking at the table above, we see that the minimum value is 1900, which is an extremely low value. This and other such values are considered antiques or just very old cars. They are outliers that could be further analyzed in a different project. So, cars with a make year earlier than 1986 are removed (2.507 records).

Finally, after dropping the outliers, there are 69.951 records left.

4.5.2 Correlation and Multicollinearity

Another important step before moving to the machine learning part is to check the relationships between the variables to understand which of them will be useful and how much they can affect the value of the dependent variable. To do that, we will check the correlation between the independent numerical variables ('odometer', 'year') and the dependent variable ('price').

As we can see in 'Figure 4,18', there is a moderate positive correlation between the price of the cars and the year at 50.67%. That's a good sign that 'year' will be very useful for creating the models. On the other hand, 'odometer' has a moderate negative correlation with the price of the cars at -44.60%. Again, 'odometer' seems useful for creating the models. This is something that we expected because, by common knowledge, the age of the car and the number of kilometers are two of the most important factors that determine a car's value.

What we also notice is that there is also a moderate negative correlation between the independent variables with -46.69%. This is something that may cause multicollinearity problems in the data. Multicollinearity may undermine the statistical significance of independent variables and make the models perform worse.

Figure 4.18: Correlation Between Numerical Variables.

To check if there is a multicollinearity problem, we can use the VIF (Variance Inflation Factor) (Aylin Alin, 2010). Using the relevant library, we can see the results below. There seems to be some multicollinearity in the year and odometer variables because the 11.08 and 6.51 values ('Figure 4.19') are a bit more than normal. Because the two values are not too high and these are two of the most important variables in the dataset, we are not going to drop any of them for now.

Figure 4.19: Multicollinearity.

	feature	VIF
0	price	3.203496
1	year	11.086593
2	odometer	6.515161

So, completing the preprocessing part, we still have 15 attributes and 69.951 records to use. Of course, more columns could have been excluded at this stage based on the results we have presented and the literature, but it is preferable to keep them all for now. We can make the final decision of which columns will be used during the model evaluation tests part.

4.5.3 Categorical Variables Encoding

The last part of preprocessing is the encoding of categorical variables. These variables, as mentioned earlier, are gathered information that is divided into groups. For example, the 'drive' variable can take the values '4wd', 'fwd', and 'rwd'. These are three different groups, specifying if a car is a four-wheel-drive, front-wheel-drive, or rear-wheel-drive.

The problem is that most machine learning algorithms cannot understand that because they only accept numerical variables as input. For them to be able to understand the categorical variables and extract their valuable information, they need to be converted into numerical ones. This process is called encoding (Guedrez et al., 2016).

There are different encoding types based on the dataset and project needs. There is generally no right or wrong method to use, but one may bring better results than the other in different situations. The encoding method we are using is called Label Encoding. In this method, each value of a categorical variable will be replaced by a numerical value. It is preferred to be used when there are ordinal

categorical variables and variables which take many different values. In our case, some categorical features are ordinal, while most of them that are not, are quite large, and one-hot encoding (another common encoding method) would lead to high memory consumption. That's why this method was chosen.

So, the categorical variables were separated, and using the LabelEncoder, they are transformed to numerical. Because all variables are now integers, 'odometer' and 'year' which were doubles, are transformed to integers too. The final dataset can be seen in 'Figure 4.20'.

Figure 4.20: Dataset After Preprocessing.

region	price	year	manufacturer	condition	cylinders	fuel	odometer	transmission	drive	size	type	paint_color	state	posting_date
16	15000	2013	12	0	5	2	128000	0	2	1	9	0	1	29
16	19900	2004	12	2	6	0	88000	0	0	1	7	1	1	25
16	14000	2012	15	0	5	2	95000	0	1	1	5	9	1	23
16	22500	2001	12	2	6	0	144700	1	2	1	9	10	1	22
16	15000	2017	9	0	6	2	90000	0	2	2	8	5	1	20

4.6 Conclusion

After extensively analyzing the dataset, we can say that it was in very good condition, related to the amount of data included. The only serious issue was the high number of NULL values which was dealt with, and some other minor issues that no longer exist. Some columns were dropped, for different reasons. Another very important thing we managed to do is get a very good knowledge of the dataset. We now know exactly what each attribute contains so we can decide how to use them. There are 15 attributes and 69.951 records.

So, now that we have both prepared the data and got a full knowledge of what it contains, we are ready to move to the machine learning part and start creating the price prediction models.

5. Model Implementation and Evaluation

5.1 Introduction

In this part, machine learning algorithms will be applied, compared, and evaluated. Machine learning is a part of AI that uses algorithms and data to copy the human thinking process and make predictions. The type of algorithms that are going to be developed are regression models. These algorithms describe the relationship between a dependent/target variable and the independent variables.

The dependent attribute, its value we will try to predict, is 'price'. For that purpose, four different types of regression models will be used. Three versions will be tested for each type, using different combinations of attributes, and their results will be evaluated to keep only the best. Finally, the different models will be compared, and the best machine learning technique will be chosen.

5.2 Chosen Programming Language and Tools

The programming language that will be used for the machine learning part is Python. As mentioned earlier, it is one of the most widely used tools in the areas of machine learning because of the high number of different functionalities it offers to the users. There are so many different libraries, making it possible to use most of the existing machine-learning algorithm types.

The libraries that will be used for the Machine Learning part are train_test_split, datasets, linear_model (LinearRegression), RandomForestRegressor, DecisionTreeRegressor, lightgbm, r2_score, mean_absolute_error, mean_squared_error and metrics. They can be spitted into three groups, based on their purpose:

- train_test_split is used to split the data to train and test.
- linear_model (LinearRegression), RandomForestRegressor, DecisionTreeRegressor, and lightgbm are the libraries used for fitting the models.
- r2_score, mean_absolute_error, mean_squared_error, and metrics are used for evaluating the results of the models.

To run the Python code, the Jupyter Notebook application will be used again, which is already installed.

5.3 Chosen Machine Learning Algorithms and Evaluation Measures

The four different algorithms which will be used are:

- Linear Regression,
- Random Forrest,
- Decision Trees, and
- Light Gradient-Boosting Machine (LightGBM).

Linear Regression is probably the most common regression model algorithm type and is used to determine the level of association between the independent and the dependent variables, using a

straight line. So, it is probably the best choice to start with. It is also good to start by using linear regression because it can show us if the variables are linearly associated.

Random Forrest is another very popular algorithm used for regression. It brings good predictions that are easily understood and can handle big datasets efficiently. This type of regression algorithm usually provides higher accuracy than others, so it is probably a good choice to use.

Decision Trees is another algorithm that is used very often for regression. It is generally easy to interpret and probably the easiest for people to understand, so it will be the third algorithm type that will be used.

Light GBM is a gradient-boosting framework that is based on decision trees. It is not as commonly used as the rest of the algorithms chosen, but it is very fast and usually achieves high accuracy performance, which can rarely be achieved in other existing boosting algorithms. So, it is a great choice to try.

After the models are created, we will need to evaluate their results and compare them. For that reason, we will need:

- R-squared,
- Mean Absolute Error and
- Root Mean Squared Error

R-squared is used to show how well the model explains observed data. It appears as a percentage, how much of the variability observed in the dependent variable is explained by the regression model. So, the higher the R-squared rate, the more capable a regression model is to explain the variability on the 'price' variable.

Mean Absolute Error shows the level of difference between the prediction of an observation and the actual value of that observation. In other words, it is the average size of the mistakes in a set of predictions.

Root Mean Squared Error measures how close a regression line is to the set of actual data points. Or the standard deviation of the residuals. Residuals are a measure showing how far from the regression line data points are. RMSE measures how spread out these residuals are. In other words, it tells us how concentrated the data is around the line of best fit.

Now that all necessary algorithms and tools that will be used have been explained, we can move to the machine learning part.

5.4 Linear Regression

We will start with linear regression. The first thing we need to do is to import the necessary libraries, which are: train_test_split, LinearRegression, r2_score, mean_absolute_error, mean_squared_error, and metrics.

5.4.1 Choosing Variables

The next step is to decide which attributes to use. After finishing the Data Analysis part, there are 15 columns left. Some of them will be more effective on the value of 'price', while some will be less.

Regarding the numerical variables this was discussed earlier, but about the categorical variables, the best way to understand which ones to use, the best way is to test different combinations and read relevant literature. Based on the literature, all the attributes we have at this time might be useful, so we will create the first linear regression model using all attributes. So, we set the target variable 'price' as y, to separate it from the rest variables, which are set as X.

5.4.2 Train

A very important step at this point is splitting the dataset into train and test data. Train data will be used to train the model, while test data will be used to evaluate the results. To do that, we are using the 'train_test_split' function, with a train/test size of 80/20 %. This is an average level of train/test to use and since we have a lot of records, we will not need a bigger train size. The 'random_state' is set to 40, to always get the same train and test data. We can now fit the train data and create our first linear regression model, using 'fit' and 'predict' functions.

5.4.3 Test

We now must check how the first model performs. To do that we will use the 'score', 'mean_absolute_error', and 'mean_squared_error' functions, to get the r-squared, mean absolute error, and root mean squared error measures values.

The value of r-squared is 56.64% which means that the model explains 56.64% of the relationship between the 'price' and independent variables. The mean absolute error is 5,021 and the root mean squared error is 6,850, which means that, related to the amount of predicted data, there is a fair number of residuals and differences between the predictions and actual values. These results indicate that the model has a fair performance but can probably be improved.

5.4.4 Experiments

We can now experiment by changing the set of variables used to train the model. For our second attempt, we will try a very different approach. While for the first model, we used all the available variables, we will now use only the ones that are considered extremely useful. These variables are 'year', 'manufacturer', condition', 'odometer', and 'type'. They were considered the most important because they are the first that a potential buyer notices on a car listing and without them, the listing would be invalid.

So, only these variables are now set as X. Data are split in train and test again using the same train/test size and 'random_state', and the train data are fitted again.

Using the same measures and functions as before, the results now are an r-squared of 31.93%, a mean absolute error of 6,516, and a root mean squared error of 8,583. This means that the second linear regression model only explains only 31.93% of the relationship between the 'price' and the five independent variables, while the number of errors and residuals has increased. So, our second attempt was not successful, as we got worse results than the first.

What we know now is that the first model using fifteen variables performed better than the second one which used only five. So, we should create a third linear regression model which will have several columns between five and fifteen, and again check if it performs better than the first.

After experimenting with many different combinations, the third model was created using twelve variables, dropping 'posting_date', 'region', and 'size'. Again, these variables are set as X and the data are split into train and test using the same train/test size and 'random_state'.

The train data are fitted again, and the results now are an r-squared of 56.59%, a mean absolute error of 5,023, and a root mean squared error of 6,854. These measures indicate that the third linear regression model performs well but is still a little worse than the first model. We can see the comparison between the three models in 'Figure 5.1'.

Figure 5.1: Results of Linear Regression Models.

Model	R-squared	Mean Absolute Error	Root Mean Squared Error
Linear Regression 1	56.641349	5021.930000	6850.550000
Linear Regression 2	31.932937	6516.500000	8583.330000
Linear Regression 3	56.597600	5023.790000	6854.000000

5.4.5 Conclusion

The first linear regression model has the highest r-squared and the lowest mean absolute error and root mean squared error values. This means that this is the best-performing linear regression model as we did not manage to improve its results. Even though, the first model's results are good and will be the one used to compare with other machine learning algorithms that will be used.

5.5 Random Forest

The second regression algorithm that we will use is random forest. Again, we need to import the necessary libraries. Some have already been imported from the linear regression part, so only RandomForestRegressor is needed.

5.5.1 Choosing Variables

We now need to decide which attributes to use. We are going to follow the same procedure with the linear regression algorithm, so we will create the first random forest regression model using all attributes. So, again, we set the target variable 'price' as y, to separate it from the rest variables, which are set as X.

5.5.2 Train

The dataset is now split into train and test data, train data to train the model, and test data to evaluate the results. To do that, we are using the 'train_test_split' function, with a train/test size of 80/20 % again. We can now fit the train data and create our first random forest regression model.

5.5.3 Test

We can now check how the first model performs. To do that we will need the 'score', 'mean_absolute_error', and 'mean_squared_error' functions, to get the r-squared, mean absolute error, and root mean squared error measures values.

The value of r-squared is 86.22% which means that the model explains 86.22% of the relationship between the 'price' and independent variables. The mean absolute error is 2,314 and the root mean squared error is 3,861, which means that, related to the amount of predicted data, there is a very low number of residuals and differences between the predictions and actual values. These results indicate that the model has a very good performance. Let's check if it can be improved further.

5.5.4 Experiments

To experiment with the variables used to create the random forest regression models, again we are going to follow the same procedure as in linear regression, and use only five variables: 'year', 'manufacturer', condition', 'odometer', and 'type'.

Only these variables are now set as X, and the data are split in train and test again using the same train/test size and 'random_state'. Then train data are fit again.

Using the same measures and functions, the results now are an r-squared of 78.94%, a mean absolute error of 2,757, and a root mean squared error of 4,773. This means that the second random forest regression model only explains only 78.94% of the relationship between the 'price' and the five independent variables, while the number of errors and residuals has increased. So, will the second attempt has returned good results, they are worse than the first.

As in the linear regression chapter, we will create a third random forest regression model which will use twelve variables, dropping 'posting_date', 'region', and 'size'. Again, these variables are set as X and the data are split into train and test using the same train/test size and 'random_state'.

The train data are fitted again, and the results we have now are an r-squared of 86.59%, a mean absolute error of 2,191, and a root mean squared error of 3,808. These measures indicate that the third random forest regression model performs very well and is even slightly better than the first one. We can see the comparison between the three models in 'Figure 5.2'.

Figure 5.2: Results of Random Forest Regression Models.

Model	R-squared	Mean Absolute Error	Root Mean Squared Error
Random Forrest 1	86.224678	2314.900000	3861.340000
Random Forrest 2	78.945642	2757.840000	4773.740000
Random Forrest 3	86.598456	2191.800000	3808.600000

5.5.5 Conclusion

The last random forest regression model has the highest r-squared and the lowest mean absolute error and root mean squared error values. This means that this is the best-performing random forest model created and that we successfully managed to improve its performance. So, It will be the one to be compared with other machine learning algorithms used.

5.6 Decision Trees

The third regression algorithm that we will use is decision trees. Once more, we need to import the necessary libraries. Some have already been imported from the previous machine learning types, so only DecisionTreeRegressor is needed.

5.6.1 Choosing Variables

Regarding which attributes to use, we are going to follow the same procedure as the previous chapters, so we will create the first decision trees regression model using all attributes. Again, we set the target variable 'price' as y, to separate it from the rest variables, which are set as X.

5.6.2 Train

We split the dataset into train and test data, train data to train the model, and test data to evaluate the results. To do that, we are using the 'train_test_split' function, with a train/test size of 80/20 % again. We can now fit the train data and create our first decision trees regression model.

5.6.3 Test

Let's see how the first model performs. We will use the 'score', 'mean_absolute_error', and 'mean_squared_error' functions again, to get the r-squared, mean absolute error, and root mean squared error measures values.

The value of r-squared is 74.45% which means that the model explains 74.45% of the relationship between the 'price' and independent variables. The mean absolute error is 2,958 and the root mean squared error is 5,257, which means that, related to the amount of predicted data, there is a low number of residuals and differences between the predictions and actual values. These results indicate that the model has a good performance, but we will try to improve it further.

5.6.4 Experiments

To experiment with the variables used to create the decision trees regression models, again we are going to follow the same procedure as in the other regression algorithm types, and use only five variables: 'year', 'manufacturer', condition', 'odometer', and 'type'.

Only these variables are now set as X, and the data are split in train and test again using the same train/test size and 'random_state'. Then train data are fit again.

Using the same measures and functions, the results now are an r-squared of 67.29%, a mean absolute error of 3,144, and a root mean squared error of 5,949. This means that the second decision trees regression model only explains only 67.29% of the relationship between the 'price' and the five

independent variables, while the number of errors and residuals has increased. So, will the second attempt has returned good results, they are worse than the first.

As in the previous chapters, we will create a third decision trees regression model which will use twelve variables, dropping 'posting_date', 'region', and 'size'. Again, these variables are set as X and the data are split into train and test using the same train/test size and 'random_state'.

The train data are fitted again, and the results we have now are an r-squared of 75.95%, a mean absolute error of 2,734, and a root mean squared error of 5,101. These measures indicate that the third model performs very well and is even better than the first one. We can see the comparison between the three models in 'Figure 5.3'.

Figure 5.3: Results of Decision Trees Regression Models.

Model	R-squared	Mean Absolute Error	Root Mean Squared Error
Decision Trees 1	74.458286	2958.000000	5257.900000
Decision Trees 2	67.292134	3144.920000	5949.950000
Decision Trees 3	75.956083	2734.770000	5101.410000

5.6.5 Conclusion

The last decision trees regression model has the highest r-squared and the lowest mean absolute error and root mean squared error values. This means that this is the best-performing decision trees model created and that we successfully managed to improve its performance. So, It will be the one to be compared with other machine learning algorithms used.

5.7 Light Gradient-Boosting Machine (LightGBM)

The last regression algorithm that we will use is light gradient-boosting machine (or LightGBM). Once more, we need to import the necessary libraries. Most have already been imported from the previous machine learning types, so only lightgbm is needed.

5.7.1 Choosing Variables

We are going to follow the same procedure as the previous chapters, so we will create the first decision trees regression model using all attributes. Again, we set the target variable 'price' as y, to separate it from the rest variables, which are set as X.

5.7.2 Train

We split the dataset into train and test data, train data to train the model, and test data to evaluate the results. To do that, we are using the 'train_test_split' function, with a train/test size of 80/20 % again. We can now fit the train data and create our first LightGBM regression model.

5.7.3 Test

Let's see how the first model performs. We will use the 'score', 'mean_absolute_error', and 'mean_squared_error' functions again, to get the r-squared, mean absolute error, and root mean squared error measures values.

The value of r-squared is 83.29% which means that the model explains 83.29% of the relationship between the 'price' and independent variables. The mean absolute error is 2,752 and the root mean squared error is 4,251, which means that, related to the amount of predicted data, there is a very low number of residuals and differences between the predictions and actual values. These results indicate that the model has a good performance, but we will try to improve it further.

5.7.4 Experiments

To experiment with the variables used to create the LightGBM regression models, we are going to follow the same procedure as in the other regression algorithm types again, and use only five variables: 'year', 'manufacturer', condition', 'odometer', and 'type'.

Only these variables are now set as X, and the data are split in train and test again using the same train/test size and 'random_state'. Then train data are fit again.

Using the same measures and functions, the results now are an r-squared of 73.05%, a mean absolute error of 3,587, and a root mean squared error of 5,400. This means that the second LightGBM regression model only explains only 73.05% of the relationship between the 'price' and the five independent variables, while the number of errors and residuals has increased. So, will the second attempt has returned good results, they are much worse than the first.

As in the previous chapters, we will create a third LightGBM model which will use twelve variables, dropping 'posting_date', 'region', and 'size'. Again, these variables are set as X and the data are split into train and test using the same train/test size and 'random_state'.

The train data are fitted again, and the results we have now are an r-squared of 83.06%, a mean absolute error of 2,772, and a root mean squared error of 4,281. These measures indicate that the third model performs very well, but slightly worse than the first one. We can see the comparison between the three models in 'Figure 5.4'.

Figure 5.4: Results of LightGBM Regression Models.

Model	R-squared	Mean Absolute Error	Root Mean Squared Error
LightGBM 1	83.298023	2752.420000	4251.790000
LightGBM 2	73.056696	3587.070000	5400.240000
LightGBM 3	83.065585	2772.590000	4281.270000

5.7.5 Conclusion

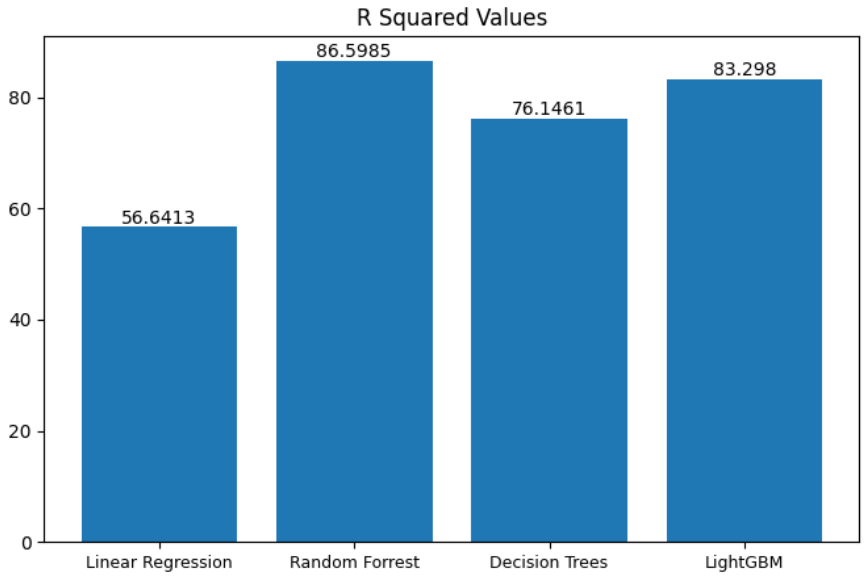
The first LightGBM regression model has the highest r-squared and the lowest mean absolute error and root mean squared error values. This means that it is the best-performing LightGBM model created and that we did not manage to improve its performance. Even though, the first model's results are very good and will be the one used to compare with other machine learning algorithms used.

5.8 Comparison of Results

At this time, three models have been implemented for each machine learning algorithm type and the best one of each has been chosen. We can now compare the results between the best models of each machine learning algorithm. To do that, we are going to compare their measures (r-squared, mean absolute error and root mean squared error) one by one and find out which one performs best.

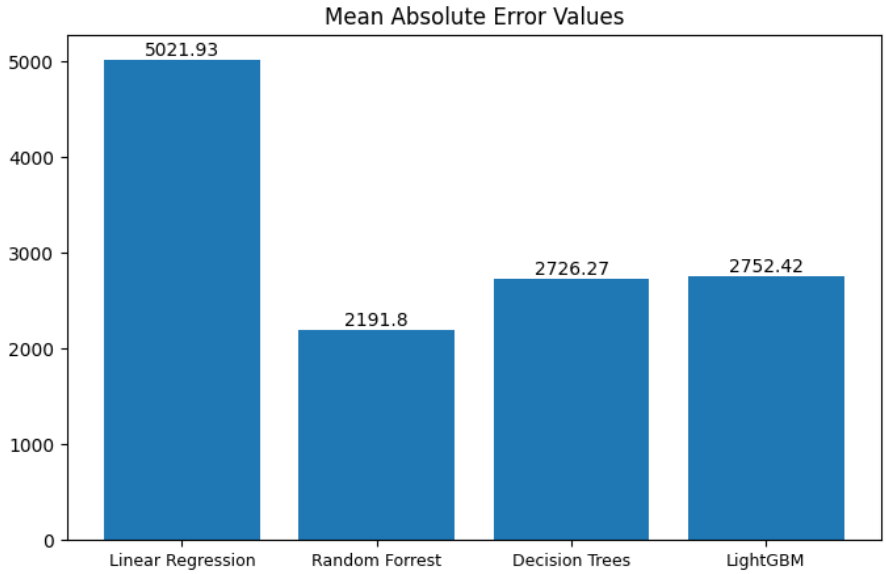
Starting with r-squared, in 'Figure 5.5' we can see that the random forest model is the one with the highest rate of about 86.6%, which is a very good result. LightGBM model also has a very good performance with about 83.3%. The third one is the decision trees model with 76.14%, which is also a good result. The last one is the linear regression model with 56.64%, which is considered a fair performance.

Figure 5.5: Values of R Squared for all model types.



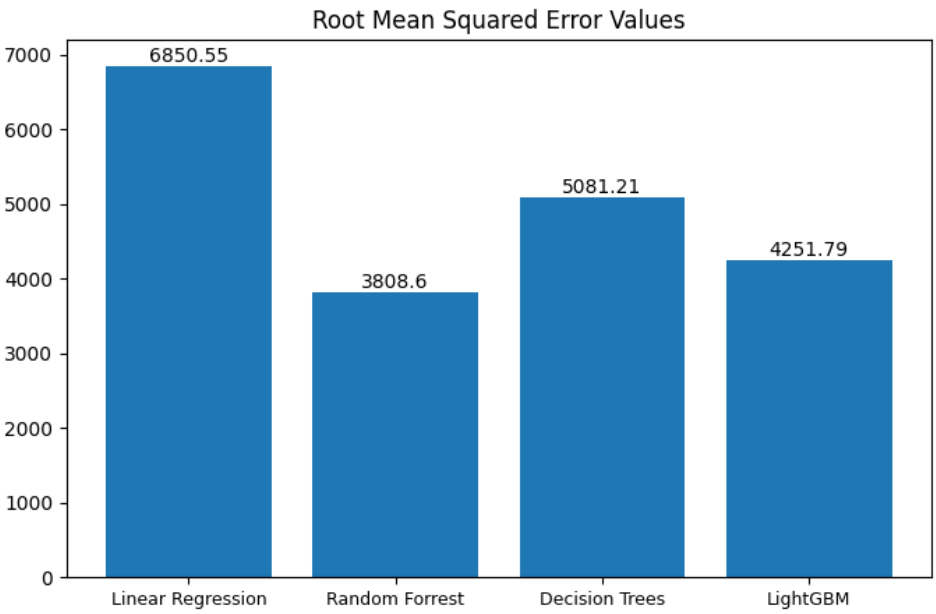
Regarding the mean absolute error levels, as we can see in 'Figure 5.6', random forest is again the one performing the best, having the lowest value of 2,191. Decision trees and LightGBM also have a good level of mean squared errors, with 2,726 and 2,752 accordingly. The linear regression model is by far the worst-performing model in terms of mean squared error values with a value of 5,021. Its performance is not so bad but related to the other machine algorithm model types, it is the worst.

Figure 5.6: Values of Mean Absolute Error for all model types.



Finally, regarding root mean squared error, as we can see in 'Figure 5.7', random forest is again the best-performing model. It has a value of only 3,808, while the second best-performing model is LIghtGBM with a value of 4,251. The third performing model is the decision trees, while the worst is again linear regression.

Figure 5.7: Values of Root Mean Squared Error for all model types.



After checking the values of the three metrics for all four model types, the random forest model is the model that performs the best out of all the models created. Looking at 'Figure 5.8', it has the highest r-squared value and the lowest mean absolute error and root mean squared error values. So, it can be used to predict car prices in the future with very good accuracy.

Figure 5.8: Results of All Regression Models.

Model	R-squared	Mean Absolute Error	Root Mean Squared Error
Linear Regression 1	56.641349	5021.930000	6850.550000
Random Forrest 3	86.598456	2191.800000	3808.600000
Decision Trees 3	76.146104	2726.270000	5081.210000
LightGBM 1	83.298023	2752.420000	4251.790000

If we wanted to rank the rest of the models, the LightGBM model is the second-best performing model, because it has a higher r-squared and the lowest root mean squared error from the rest. Its mean squared error value is slightly higher than the decision trees model, but it is very close.

The third model in the ranking is the decision trees and the last one is the linear regression. The last one is by far the worst-performing model out of the four, having both the lowest r-squared value and the highest mean absolute error and root mean squared error values.

5.9 Conclusion

We can say that the model implementation and evaluation part was successful. We created several models using different machine learning algorithm types, evaluated their results, and kept only one of each type. Finally, after comparing the models of different algorithm types, we ended up ranking them based on their performance.

Random forest was the best was the best-performing algorithm, followed closely by LightGBM. The third best model type was the decision trees, while linear regression had by far the worst performance. All of them can be used for predicting a used car's price, but the random forest is the one we suggest because it has the highest accuracy and the lowest number of errors.

6. Conclusion

6.1 Summary

To conclude, we managed to create a random forest regression model that can predict the price of used cars with good accuracy. To achieve this, we used the raw dataset and first used some exploratory data analysis to understand what kind of information it contains. Then we did some preprocessing of the data to make some necessary changes and complete the data analysis. After that, we moved to the model implementation part, where four different regression algorithm types were used to train and test three different models for each type. The results of each model type were evaluated separately, to identify the best-performing model for each regression type. Finally, the four best models' results were compared, and the best model was chosen.

6.2 Evaluation of Project Requirements

The main requirement of this project was to create a regression model that can predict the price of used cars. This was achieved, and the performance of the final model is very good. To reach that model, it was also required to create multiple models with different regression algorithm types and compare their results. This process was done successfully because many models were created, most of which had a good performance. All of them were compared to identify the best one.

Another requirement of the project, before creating any models, was to perform data analysis on the dataset. Some exploratory analysis was done to understand the dataset and preprocessing to prepare the data for the modelling part. So, all the requirements were achieved.

6.3 Project Management

Regarding project management, almost everything went according to the initial planning. All the subtasks were completed as planned, so the final project was also completed on time. There was only one change that occurred in the 'Model Implementation and Evaluation' part.

In the initial planning phase of this chapter, where the models had to be created, it was decided that only three regression model types would be used. Linear regression, random forest, and decision trees. This decision changed while doing some research on how to choose the most appropriate algorithm types for machine learning and on what other algorithms could be used. At that point, I learned more about LightGBM regressor and decided to add it as an extra algorithm to the project. This addition was done because the rest algorithm types are some of the most common ones for such projects, while LightGBM is used more rarely. It is also very powerful, and this could result in creating a model with higher performance. So, this change offered more variety to the models that were created and some very high-performing regression models.

6.4 Future Recommendation

Future recommendations about improving or expanding the results of this project refer to improving the quality of the created models and increasing the accuracy of the best model. There are some recommendations which maybe could achieve that.

The first recommendation is to try using different regression algorithm types. Of course, the results were very satisfying, but there are several other algorithms that can be used for the same purpose and maybe some of them could achieve even higher performance.

Another recommendation is to make changes to the preprocessing part. The way that the data are handled and manipulated can affect the model's performance. So, by making changes to the dataset, the performance of the models could possibly be improved.

My last recommendation is to apply the same or a similar process to a different dataset. The dataset used in this project was very high quality, but there is always room for improvement. A different dataset would have other attributes which contain other information. This information could help increase the model's performance.

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Appendix 1: Project Plan

Project Plan														
Project Activities	25-Feb	11-Mar	25-Mar	08-Apr	22-Apr	06-May	20-May	03-Jun	17-Jun	01-Jul	15-Jul	29-Jul	12-Aug	25-Aug
Project Idea														
Project Specification														
Research														
Data Analysis														
Modeling														
Evaluation														
Write-up														
Final Project Demo														
Hand-in														

Appendix 2: Project Specification

UNIVERSITY OF PORTSMOUTH

School of Computing Postgraduate Programme

MSc in Data Analytics *(online)*

Project Specification

DIMITRIOS THEOCHARIS

Project Specification

1. Basic details

Student name:	Dimitrios Theocharis
Draft project title:	Car Prices Analysis and Prediction Using Machine Learning
Course and year:	MSc Data Analytics Dissertation Project 2023
Client organisation:	University of Portsmouth
Client contact name:	Ramazan Esmeli
Project supervisor:	Ramazan Esmeli

2. Outline of the project environment

The client is my supervisor Ramazan Esmeli and he needs machine learning models on predicting car prices and the performance analysis of the models to find which one performs best.

3. The problem to be solved

The aim of the project is to analyse past car sales in U.S.A and create models than can predict the price of used cars and compare how accurate they are.

The constraints are that there is a big variety of car makes in the market, producing different kinds of cars. Another constraint is choosing the ideal predictive model by considering the available features of the dataset that would be limiting the models' performance. Also a serious constraint is the limited time to submit the project.

4. Breakdown of tasks

I am going to use an open-source dataset I found on the internet which contain data about used car sales. I am going to analyse the data to get to know the information of the data and understand the problem better. Then I will move with creating predictive models about the cars' prices.

Some programming languages or tools like Python, Pyspark, SQL and Tableau will be used.

I will need to perform an extended data analysis, make visualisations and create regression models. I will acquire any skills needed using the help of online websites, books and my university teachers.

Project Plan														
Project Activities	25-Feb	11-Mar	25-Mar	08-Apr	22-Apr	06-May	20-May	03-Jun	17-Jun	01-Jul	15-Jul	29-Jul	12-Aug	25-Aug
Project Idea														
Project Specification														
Research														
Data Analysis														
Modeling														
Evaluation														
Write-up														
Final Project Demo														
Hand-in														

5. Project deliverables

A complete data analysis, including statistics, graphs, and charts, and some regression models. The final documents (apart from the report) will be the code used for the analysis and modelling.

6. Requirements

Machine learning models that can estimate the car price which have different specifications.

A dissertation report that outlines clearly the motivation, objective and applied methodology, experiments, results and conclusion.

First, I will have to explore what are the recent publications on this topic. Then I will do data analysis to understand the dataset and build the machine learning models.

7. Legal, ethical, professional, social issues

I won't have any ethical issues because I am using a publicly available dataset. An ethics review will be performed through the University of Portsmouth website.

8. Facilities and resources

I am going to use my own computer, and software provided by the university or free software.

9. Project plan

Until March 25, I had already found my complete project idea and completed the project specification. The next step is doing some research to get ideas and be ready to start working on the project. Then, I am going to start working on the Data Analysis of the dataset by the end of May. During June and early July, I am going to work on the models and their evaluation. During all this time and until the middle of August, I will have finished writing the dissertation report. In the next few days, I will do the presentation, and until August 25 I will hand in the dissertation project.

10. Supervision meetings

We are going to have meetings once or twice every two weeks via online video calls, based on the stage of the project.

11. Project mode

	Please delete as appropriate	
Registration mode		Part Time
Project mode		Part Time
Planned submission deadline	25 August 2023	

12. Signatures

	Signature:	Date:
Student	Dimitrios Theocharis(DT)	24 March 2023
Client	Ramazan Esmeli(RE)	23 March 2023
Project supervisor	Ramazan Esmeli(RE)	23 March 2023

Appendix 3: Ethics Review of Project

noreply@port.ac.uk
to me

Thank you for using the online ethics Ethics Screening tool

You have indicated that your study does not include any of the following:

- Human participants (taking tests, being observed, answering questionnaires, taking part in interviews/focus groups etc.)
- Gathers or uses confidential information that might identify human participants
- Includes "Relevant material" as defined by the Human Tissue Act 2004
- Includes animals (and you do not have permission from the University's AWERB committee to proceed)
- Has an environmental impact
- Impacts our cultural heritage (excavation, destructive sampling etc.)
- Requires review from an external ethics committee (NHS, MOD, PHE, HMPPS etc.)
- Has health and safety concerns that cannot be met by normal risk assessment

If this is correct then please use this email as evidence of ethics review. Your reference number is **ETHICS-10664.**

If, however, you think your study will after-all involve any of the above please refer to http://www2.port.ac.uk/research/ethics/